

22050 Continuous-Time Signals and Linear Systems

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L08

Forward and Inverse Laplace Transformation Laplace Theorems

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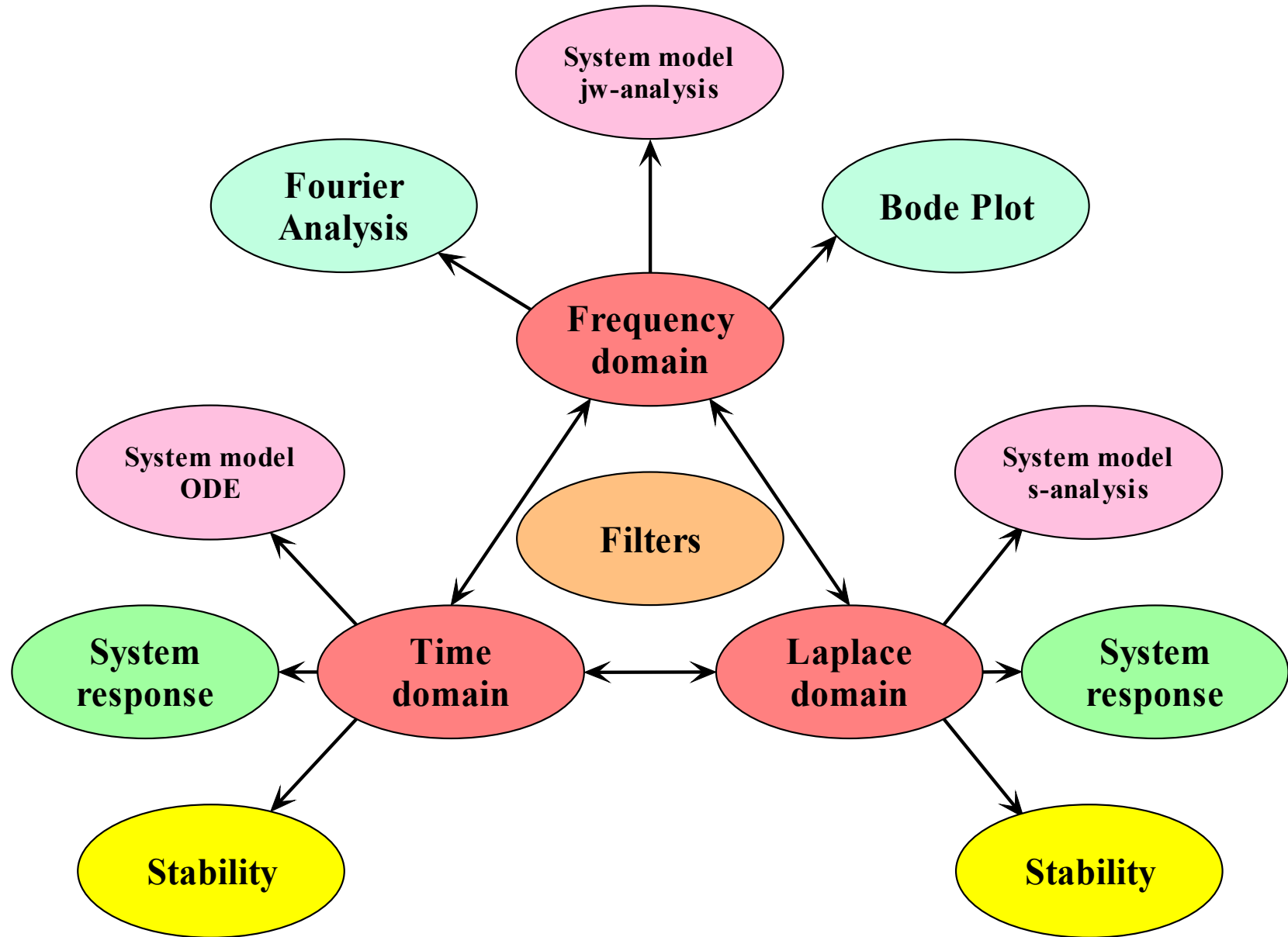
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- System
 - Model
 - Response
 - Performance
- Domains
 - Time
 - Frequency
 - s-domain



6.1 Laplace Transform

6.2 Some properties of the Laplace Transform

The Fourier Transformation has two important limitations:

1. Convergence:
2. It cannot handle transient behaviors due to the abrupt onset of a signal. Fourier transformation does not naturally incorporate initial conditions in system descriptions.

$$\int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt < \int_{-\infty}^{\infty} |x(t)| |e^{-j\omega t}| dt = \int_{-\infty}^{\infty} |x(t)| dt < \infty$$

Solution:

As far as convergence is concerned, we can modify the signal in order to ensure convergence of the Fourier integral. We do this by damping the signal adequately with a decaying exponential:

Then we can always find a value σ that makes $|\varphi(t)|$ integrable.

$$\varphi(t) = e^{-\sigma t} \cdot x(t)u(t)$$

Provided that: $x(t)$ is an exponential order signal

$$|x(t)| < Me^{\alpha t}u(t), 0 < M, \alpha < \infty$$

Example

$$x(t) = e^{3t}u(t)$$

$$\varphi(t) = e^{-\sigma t} \cdot e^{3t}u(t) = e^{(3-\sigma)t}u(t)$$

$$3 - \sigma < 0 \Rightarrow \sigma > 3$$

Fourier Transform of exponentially damped signal

Forward transform:

$$X(j\omega) \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

What transform do we get if we multiply a time function by an exponentially decaying function?

$$\begin{aligned} \mathcal{F}\{e^{-\sigma t} x(t)\} &= \underbrace{\int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt}_{\text{Fourier transform}} = \underbrace{\int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt}_{\text{Laplace transform}} \\ &= X(\sigma + j\omega) \end{aligned}$$

The transform is a function of a complex-valued frequency variable.

We call this the **forward Laplace transform**.

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad s \stackrel{\text{def}}{=} \sigma + j\omega$$

We started out with a Fourier transformation of a product $e^{-\sigma t} x(t)$. Then we reinterpret the expression as a new type of transformation of $x(t)$ where the pre-multiplication by $e^{-\sigma t}$ is an integral part of the transformation rather than a part of the signal being transformed. In doing so, we observe that the Laplace transform is equipped to handle exponentially rising signals, - provided that we choose σ such that product $e^{-\sigma t} x(t)$ has a Fourier transformation. This simply implies that the product $e^{-\sigma t} x(t)$ must be a decaying signal in time, i.e., that $e^{-\sigma t}$ falls at a faster rate than $x(t)$ increases. If $x(t)$ is an exponential order signal such a σ always exists.

Fourier Transform of exponentially damped signal

This is the forward Fourier transform of the function $x(t)e^{-\sigma t}$:

$$x(t)e^{-\sigma t} \leftrightarrow \int_{-\infty}^{\infty} x(t)e^{-\sigma t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t)e^{-(\sigma+j\omega)t} dt = X(\sigma + j\omega)$$

We can then write the inverse Fourier transform:

$$x(t)e^{-\sigma t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega) e^{j\omega t} d\omega$$

Multiplying on both sides with $e^{\sigma t}$:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega) e^{(\sigma+j\omega)t} d\omega$$

Variable substitution: $s \stackrel{\text{def}}{=} \sigma + j\omega \Rightarrow ds = jd\omega$

Inverse Laplace Transform:

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{s t} ds$$

Complex frequency: $s \stackrel{\text{def}}{=} \sigma + j\omega$ $ds = j \cdot d\omega$

In the s -plane, we integrate from $\sigma - j\infty$ to $\sigma + j\infty$

Region of convergence: What should be the value of σ ?
It should have a value c such that $|x(t)e^{-ct}|$ is integrable

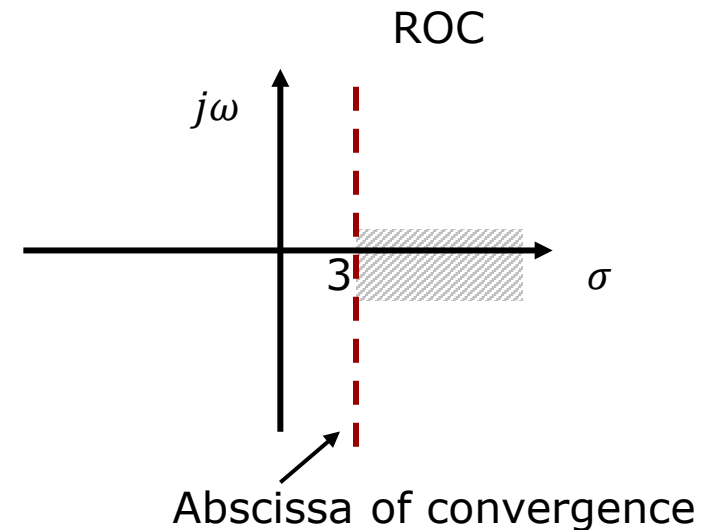
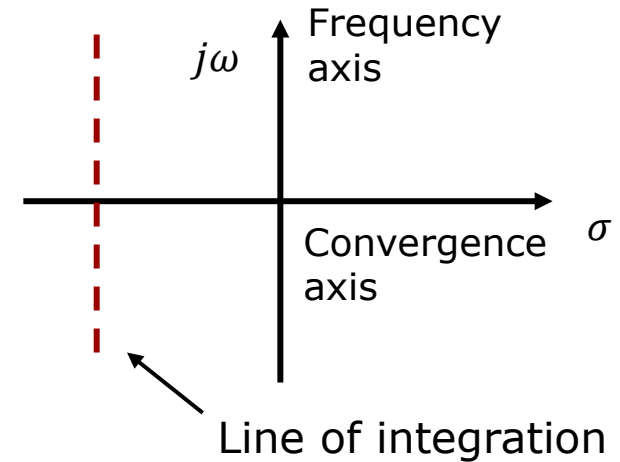
Example: $x(t) = e^{3t}, t > 0$

We must now determine c , so that: $|e^{-ct} \cdot e^{3t}| = |e^{(3-c)t}|$

$$c > 3$$

is integrable. This is the case when:

The region of convergence (ROC) for $\mathcal{L}\{e^{3t}u(t)\}$ is $\sigma > 3$.



Forward transformation:

$$\begin{aligned}\mathcal{F}\{e^{-\sigma t}x(t)\} &= \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt \\ &= X(\sigma + j\omega)\end{aligned}$$

$$s \stackrel{\text{def}}{=} \sigma + j\omega$$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$X(s) \stackrel{\text{def}}{=} \mathcal{L}\{x(t)\}$$

Transform pair:

$$X(s) \leftrightarrow x(t)$$

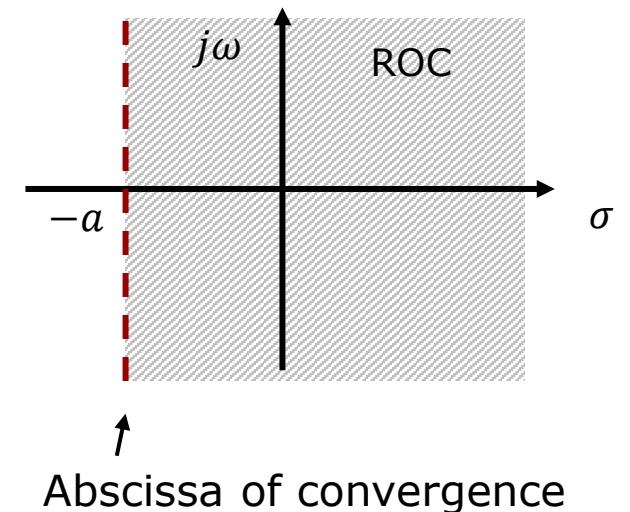
Example

$$x(t) = e^{-at}u(t) \quad a \in \mathbb{R}$$

$$X(s) = \int_0^{\infty} e^{-at}e^{-st}dt = \int_0^{\infty} e^{-(s+a)t}dt = -\frac{1}{s+a}e^{-(s+a)t}\bigg|_0^{\infty} = \frac{1}{s+a} - \lim_{t \rightarrow \infty} \frac{1}{s+a}e^{-(s+a)t}$$

$$\lim_{t \rightarrow \infty} e^{-(s+a)t} = \lim_{t \rightarrow \infty} e^{-\text{Re}(s+a)t - j\text{Im}(s+a)t} = \lim_{t \rightarrow \infty} \underbrace{e^{-\text{Re}(s+a)t}}_{\text{modulus}} e^{-j\omega t} = \begin{cases} 0 & \text{Re}(s+a) > 0 \\ \infty & \text{Re}(s+a) < 0 \end{cases}$$

$$X(s) = \frac{1}{s+a}, \text{Re}(s+a) > 0 \quad \text{Re}(s) > -a$$



Uniqueness of bilateral Laplace Transform

Without stating the region of convergence, the bilateral transformation is not unique, as the following example will show:

$x(t)$ causal

$a \in \mathbb{R}$

$$x(t) = e^{-at}u(t)$$

$$X(s) = \frac{1}{s+a}, \text{Re}\{s\} > -a$$

$g(t)$ anti-causal

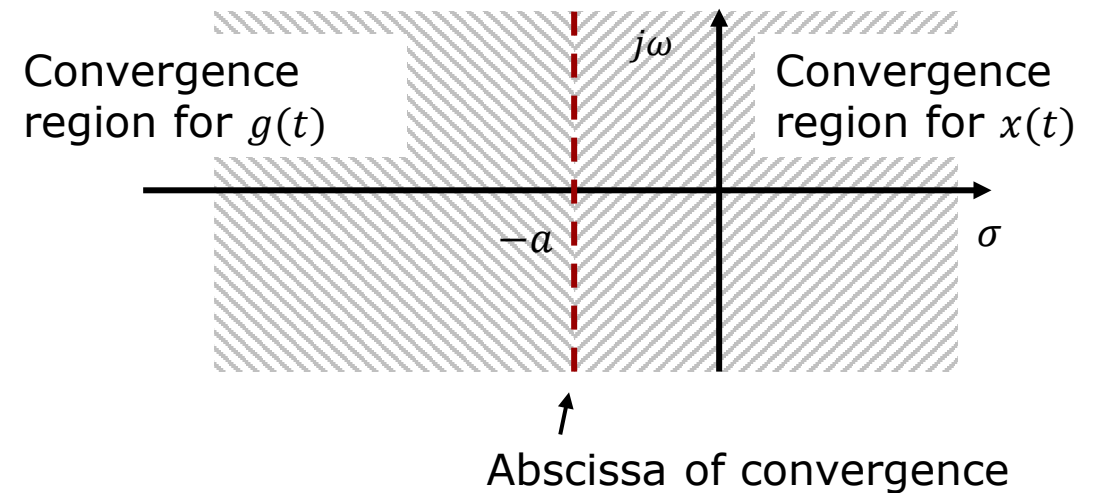
$a \in \mathbb{R}$

$$g(t) = -e^{-at}u(-t)$$

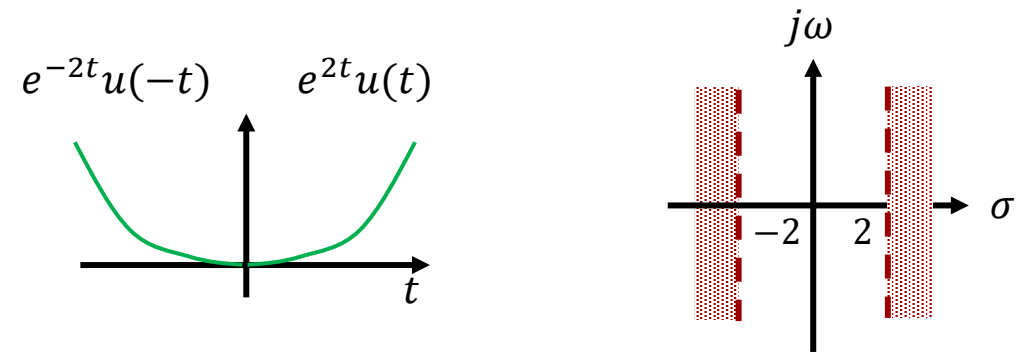
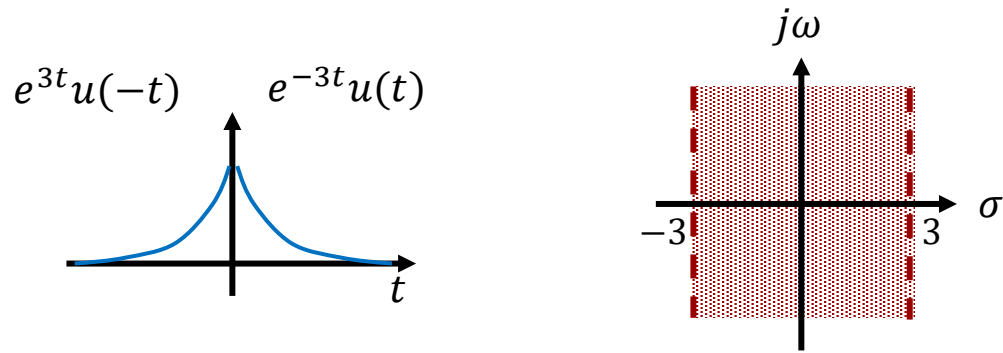
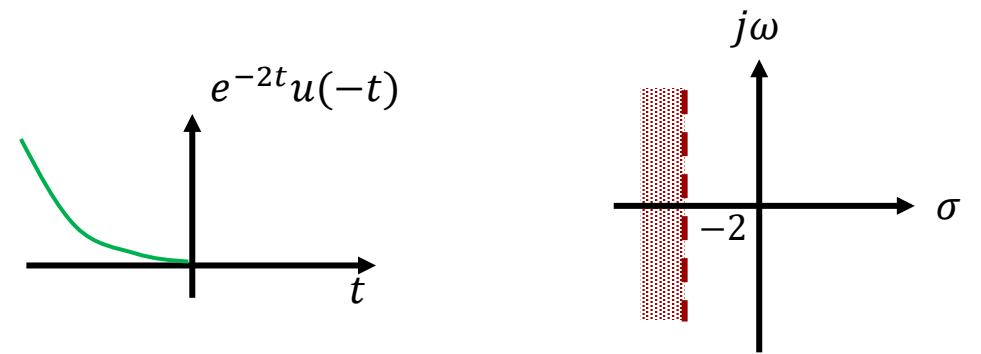
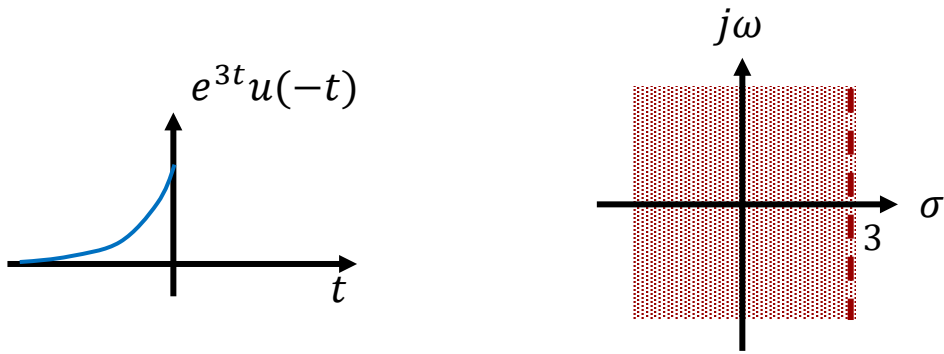
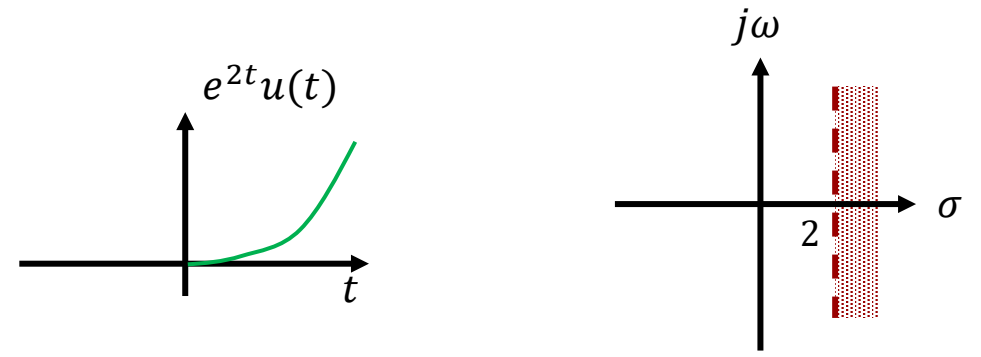
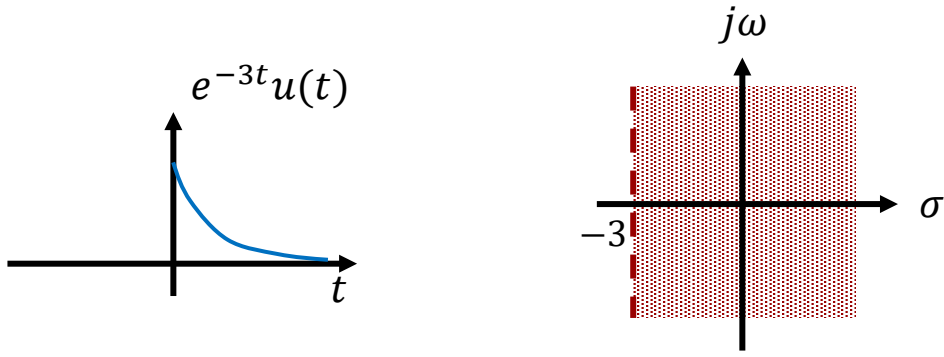
$$G(s) = \frac{1}{s+a}, \text{Re}\{s\} < -a$$

The bilateral Laplace Transform is only unique when expressed together with the region of convergence.

In this example the two functions have the same Laplace transform. The only difference is the region of convergence.



Regions of convergence



All causal signals have convergence regions extending to the **right** from a vertical line c .

All anti-causal signals have convergence regions extending to the **left** from a vertical line c .

If we from hereon decide that we only consider causal signals, then the convergence region is given, and the Laplace transformation is unique.

The Laplace transformation of causal signals is called the **UNILATERAL Laplace transformation**.

In Signals and Linear Systems, the unilateral Laplace transform is by far the most used Laplace transform, and without other indications:

the term "Laplace transform" is in this course taken to mean "the unilateral Laplace transform".

Remember this at the exam.

$$X(s) = \int_{0-}^{\infty} x(t)e^{-st} dt$$

$$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st} ds$$

The unilateral Laplace transform puts a constraint on $x(t)$, not $X(s)$. Hence, the inverse Laplace transform is the same.

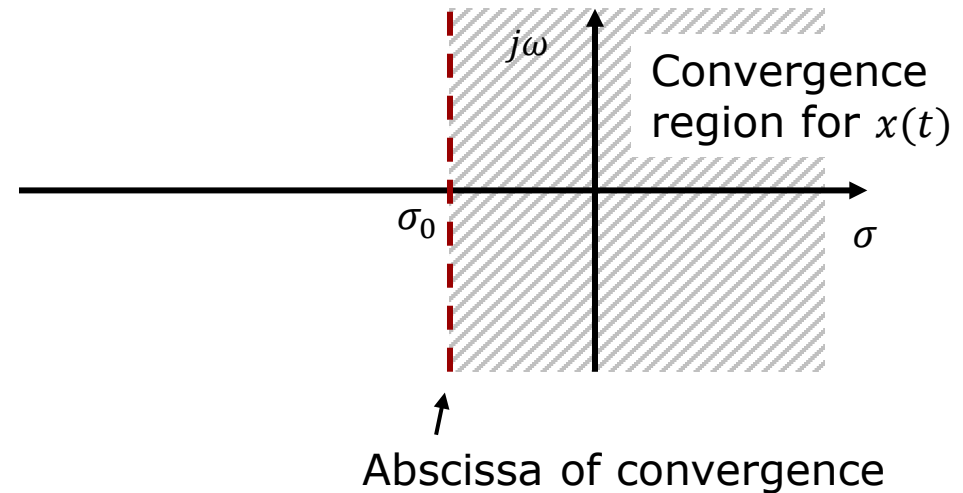
Existence of Unilateral Laplace Transformation

If $x(t)$ is an exponential order signal, i.e.,

If: $|x(t)| < M e^{\sigma_0 t}$ and $\sigma > \sigma_0$ then

$$\begin{aligned} X(s) &= \int_{0_-}^{\infty} x(t) e^{-st} dt = \int_{0_-}^{\infty} [x(t) e^{-\sigma t}] e^{-j\omega t} dt \\ &< \int_{0_-}^{\infty} |[x(t) e^{-\sigma t}] e^{-j\omega t}| dt = \int_{0_-}^{\infty} |x(t) e^{-\sigma t}| dt < \int_{0_-}^{\infty} |M e^{\sigma_0 t} e^{-\sigma t}| dt < M \int_{0_-}^{\infty} |e^{-(\sigma - \sigma_0) t}| dt < \infty \end{aligned}$$

If $x(t)$ is of exponential order σ_0 , then the unilateral Laplace transform converges for $\Re(s) > \sigma_0$.

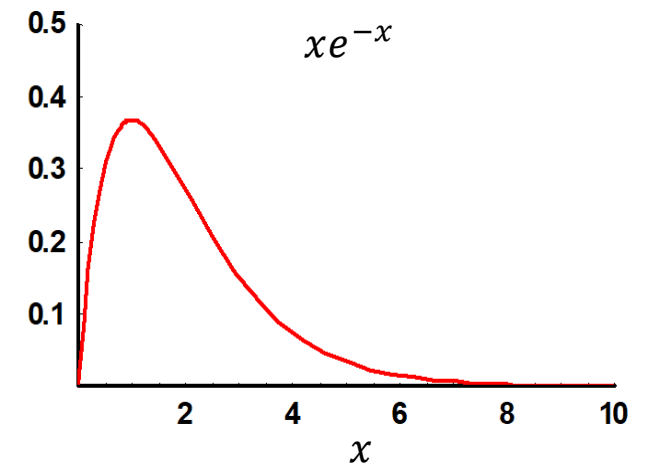


$$x(t) = \delta(t) \quad X(s) = \int_{0^-}^{\infty} \delta(t) e^{-st} dt = \int_{0^-}^{\infty} \delta(t) e^{-s \cdot 0} dt = \int_{0^-}^{\infty} \delta(t) dt = 1$$

$$x(t) = u(t) \quad X(s) = \int_{0^-}^{\infty} u(t) e^{-st} dt = \int_{0^+}^{\infty} e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_{0^+}^{\infty} = \frac{1}{s}, \operatorname{Re}(s) > 0$$

$$\begin{aligned} x(t) = tu(t) \quad X(s) &= \int_{0^-}^{\infty} tu(t) e^{-st} dt = \int_{0^+}^{\infty} te^{-st} dt = -\frac{1}{s^2} e^{-st} (1 + st) \Big|_{0^+}^{\infty} \\ &= \frac{1}{s^2} - \lim_{t \rightarrow \infty} \frac{(1 + st)e^{-st}}{s^2} \\ &= \frac{1}{s^2}, \operatorname{Re}(s) > 0 \end{aligned}$$

The limit of
 $st e^{-st}$



Comparison with Fourier transform

$x(t)$	$X(\omega)$	$X(s)$
$\delta(t)$	1	1
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$	$\frac{1}{s}, \sigma > 0$
$tu(t)$	nonexistent	$\frac{1}{s^2}, \sigma > 0$

For the unit step function, the region of convergence does not include the imaginary axis. This implies that the unit step function does not have a frequency characteristic in the ordinary sense.

We recall from the Fourier transformation of the step function, that the frequency characteristic exists only in a generalized sense.

The class of signals with a Laplace transform is larger than that with a Fourier transform.

For the inverse Laplace transform, there is no difference between the bilateral transform and the unilateral transform.

$$e^{-\sigma t} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega) e^{j\omega t} d\omega$$

$$\Updownarrow$$

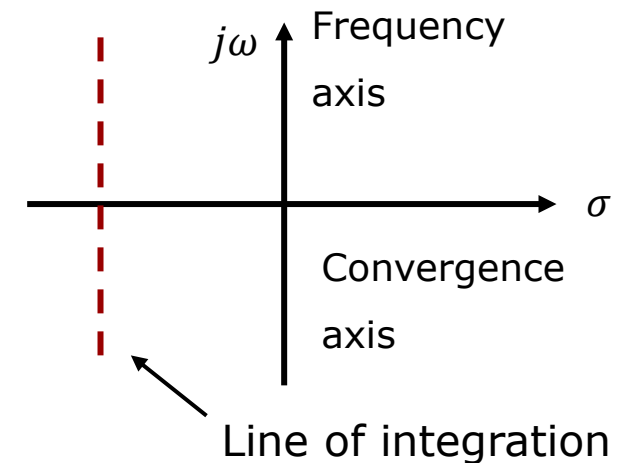
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega) e^{(\sigma + j\omega)t} d\omega$$

In the s-plane we integrate from $\sigma - j\infty$ to $\sigma + j\infty$

$$s \stackrel{\text{def}}{=} \sigma + j\omega$$

$$ds = j \cdot d\omega$$

$$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s) e^{st} ds$$



In general, the computation of this integral requires integration in the complex plane.

$$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st} ds$$

To avoid this, the function $X(s)$ is factorized in simple factors and **partial fraction expansion** is performed.

$$\begin{aligned} X(s) &= \frac{P(s)}{Q(s)} = \frac{(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} \\ &= \frac{A_1}{s + p_1} + \frac{A_2}{s + p_2} + \cdots + \frac{A_n}{s + p_n} \end{aligned}$$

Partial Fraction Expansion
(Lathi appendix B.5)

Using **tables of Laplace transforms**, we then inverse transform term by term:

$$x(t) = [A_1 e^{-p_1 t} + A_2 e^{-p_2 t} + \cdots + A_n e^{-p_n t}] u(t)$$

Partial fraction expansion

Video 2

Let us consider the step response:

$$y(t) = h(t) * x(t) = Ae^{-at} * u(t)$$

It will be shown in a later slide that convolution in the time domain corresponds to multiplying the transforms in the Laplace domain:

$$Y(s) = H(s) \cdot X(s) = \frac{A}{s+a} \cdot \frac{1}{s} = \frac{A}{s(s+a)}$$

To obtain the solution in the time domain, we need to find the inverse Laplace transform of $Y(s)$. This is done by splitting up the expression into several fractions and then finding the inverse Laplace transform of each term.

$$y(t) = \mathcal{L}^{-1} \left[\frac{A}{s(s+a)} \right] = \mathcal{L}^{-1} \left[\frac{k_1}{s} + \frac{k_2}{s+a} \right] = \mathcal{L}^{-1} \left[\frac{k_1}{s} \right] + \mathcal{L}^{-1} \left[\frac{k_2}{s+a} \right]$$

The technique of splitting a fraction up into simpler fractions is called **Partial Fraction Expansion**.

To determine the unknown coefficient k_1 , we need to isolate it.

We do this by multiplying both sides of the equation with its denominator (s) and then setting $s = 0$. This removes the term with k_2 .

$$Y(s) = \frac{A}{s(s+a)} = \frac{k_1}{s} + \frac{k_2}{s+a}$$

$$s \cdot Y(s) \Big|_{s=0} = s \cdot \frac{A}{s(s+a)} \Big|_{s=0} = \underset{\text{green}}{s} \cdot \frac{k_1}{\underset{\text{green}}{s}} \Big|_{s=0} + \underset{\text{red}}{s} \cdot \frac{k_2}{\underset{\text{red}}{s+a}} \Big|_{s=0}$$

$$s \cdot Y(s) \Big|_{s=0} = \frac{A}{(0+a)} = k_1 + \underset{\text{red}}{0} \cdot \frac{k_2}{a}$$

$$k_1 = \frac{A}{a}$$

Next, we use the same approach to isolate k_2 :

$$\begin{aligned}(s + a) \cdot Y(s) \Big|_{s=-a} &= (s + a) \cdot \frac{A}{s(s + a)} \Big|_{s=-a} \\ &= (s + a) \cdot \frac{k_1}{s} \Big|_{s=-a} + (s + a) \cdot \frac{k_2}{s + a} \Big|_{s=-a}\end{aligned}$$

$$(s + a) \cdot Y(s) \Big|_{s=-a} = \frac{A}{s} \Big|_{s=-a} = 0 \cdot \frac{k_1}{s} \Big|_{s=-a} + k_2 \Rightarrow k_2 = -\frac{A}{a}$$

Inserting the two coefficients, we get:

$$Y(s) = \frac{A}{s(s + a)} = \frac{k_1}{s} + \frac{k_2}{s + a} = \frac{A}{a} \cdot \frac{1}{s} - \frac{A}{a} \cdot \frac{1}{s + a}$$

Looking up the inverse Laplace transforms of each term, we get:

$$\begin{aligned}y(t) &= \frac{A}{a} \cdot \mathcal{L}^{-1} \left[\frac{1}{s} \right] - \frac{A}{a} \cdot \mathcal{L}^{-1} \left[\frac{1}{s + a} \right] = \frac{A}{a} \cdot u(t) - \frac{A}{a} \cdot e^{-at} \cdot \underbrace{u(t)}_{\text{for causality}} \\ &= \frac{A}{a} (1 - e^{-at}) \cdot u(t)\end{aligned}$$

Here is a problem
with repeated
roots:

$$X(s) = \frac{A}{s(s+a)^2} = \frac{k_0}{s} + \frac{k_1}{(s+a)} + \frac{k_2}{(s+a)^2}$$

$$k_0 = s \frac{A}{s(s+a)^2} \Big|_{s=0} - s \frac{k_1}{(s+a)} \Big|_{s=0} - s \frac{k_2}{(s+a)^2} \Big|_{s=0} = \frac{A}{a^2}$$

$$k_2 = (s+a)^2 \frac{A}{s(s+a)^2} \Big|_{s=-a} - (s+a)^2 \frac{k_0}{s} \Big|_{s=-a} - (s+a)^2 \frac{k_1}{(s+a)} \Big|_{s=-a} = -\frac{A}{a}$$

$$X(s) = \frac{A}{s(s+a)^2} = \frac{A}{a^2} \frac{1}{s} + \frac{k_1}{(s+a)} - \frac{A}{a} \frac{1}{(s+a)^2}$$

Set $s = 1$ and solve for k_1 :

$$X(s) = \frac{A}{s(s+a)^2} = \frac{A}{a^2} \frac{1}{s} + \frac{k_1}{(s+a)} - \frac{A}{a} \frac{1}{(s+a)^2}$$

$$X(1) = \frac{A}{1(1+a)^2} = \frac{A}{a^2} \frac{1}{1} + \frac{k_1}{(1+a)} - \frac{A}{a} \frac{1}{(1+a)^2}$$

$$X(1) = \frac{A}{(1+a)^2} = \frac{A}{a^2} + \frac{k_1}{(1+a)} - \frac{A}{a} \frac{1}{(1+a)^2}$$

$$\frac{k_1}{(1+a)} = \frac{A}{(1+a)^2} + \frac{A}{a} \frac{1}{(1+a)^2} - \frac{A}{a^2}$$

$$\begin{aligned} k_1 &= \frac{A}{(1+a)} + \frac{A}{a} \frac{1}{(1+a)} - \frac{A(1+a)}{a^2} = A \left(\frac{a^2}{a^2(1+a)} + \frac{a}{a^2(1+a)} - \frac{(1+a)^2}{a^2(1+a)} \right) \\ &= A \left(\frac{a^2 + a - (1 + 2a + a^2)}{a^2(1+a)} \right) = A \left(\frac{-1 - a}{a^2(1+a)} \right) = -\frac{A}{a^2} \end{aligned}$$

$$X(s) = \frac{A}{s(s+a)^2} = \frac{A}{a^2} \left[\frac{1}{s} - \frac{1}{(s+a)} - \frac{a}{(s+a)^2} \right]$$

Inverse Laplace transformation of each term using a table of transforms:

$$x(t) = \frac{A}{a^2} [1 - e^{-at} - at \cdot e^{-at}] \cdot u(t)$$

Let us consider the step response of a second order lowpass filter.

We will discover that the Laplace transform of the impulse response is:

We will often prefer to express it in terms of the **resonance frequency** ω_n and its **damping factor** ζ :

To obtain the solution in the time domain, we need to inverse Laplace transform this:

This requires partial fraction expansion.

$$y(t) = h(t) * u(t)$$

$$H(s) = \frac{b_0}{s^2 + a_1s + a_0}$$

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2} \quad \omega_n = \sqrt{a_0} \quad \zeta = \frac{a_1}{2\sqrt{a_0}}$$

$$Y(s) = H(s) \cdot \frac{1}{s}$$

$$Y(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_ns + \omega_n^2)} = \frac{k_0}{s} + \frac{k_1s + k_2}{s^2 + 2\zeta\omega_ns + \omega_n^2}$$

$$Y(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{k_0}{s} + \frac{k_1 s + k_2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Solving for k_0 :

$$k_0 = s \cdot Y(s) \Big|_{s=0} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \Big|_{s=0} = 1$$

Solving for k_1 :

$$s \cdot Y(s) \Big|_{s \rightarrow \infty} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \Big|_{s \rightarrow \infty} = \frac{k_0 s}{s} \Big|_{s \rightarrow \infty} + \frac{k_1 s^2}{s^2} \Big|_{s \rightarrow \infty} = k_0 + k_1 = 0 \quad k_1 = -k_0 = -1$$

Solving for k_2 :

$$s = 1 \Rightarrow \frac{\omega_n^2}{1(1^2 + 2\zeta\omega_n 1 + \omega_n^2)} = \frac{1}{1} + \frac{-1 \cdot 1 + k_2}{1^2 + 2\zeta\omega_n 1 + \omega_n^2} \Rightarrow k_2 = -2\zeta\omega_n$$

Inserting results:

$$Y(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Techniques of partial fraction expansions

Next, we need to use a table of transforms. This often requires switching to other symbols:

$$Y(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s} - \frac{As + B}{s^2 + 2as + c}$$

$$y(t) = [1 - re^{-at} \cos(bt + \theta)]u(t)$$

Solution in table of transforms

$$r = \sqrt{\frac{A^2c + B^2 - 2ABa}{c - a^2}} = \frac{1}{\sqrt{1 - \zeta^2}}$$

$$\theta = \tan^{-1}\left(\frac{Aa - B}{A\sqrt{c - a^2}}\right) = \tan^{-1}\left(\frac{-\zeta}{\sqrt{1 - \zeta^2}}\right)$$

$$b = \sqrt{c - a^2} = \omega_n\sqrt{1 - \zeta^2}$$

To obtain the solution using our notation, some work is required to rephrase the solution:

$$y(t) = \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \cos\left(\omega_n\sqrt{1 - \zeta^2}t + \tan^{-1}\left[\frac{-\zeta}{\sqrt{1 - \zeta^2}}\right]\right)\right]u(t)$$

Geometric relations of poles

We would like to find the poles:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s_{1,2} = -\zeta\omega_n \pm j \cdot \omega_n\sqrt{1-\zeta^2}$$

$$\omega_n = |s_1|$$

$$\cos \theta = \frac{\zeta\omega_n}{\omega_n} = \zeta \Rightarrow \theta = \cos^{-1} \zeta$$

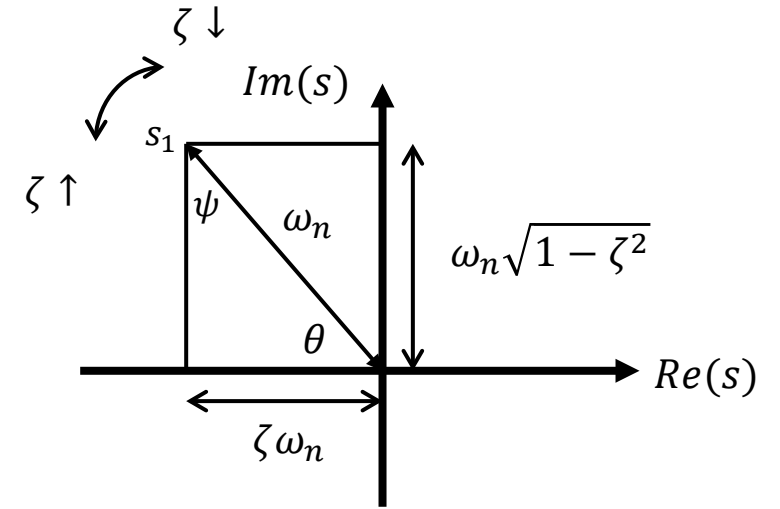
$$\tan \psi = \frac{\zeta\omega_n}{\omega_n\sqrt{1-\zeta^2}} = \frac{\zeta}{\sqrt{1-\zeta^2}} \Rightarrow$$

$$\psi = \tan^{-1} \frac{\zeta}{\sqrt{1-\zeta^2}}$$

$$\psi = 90^\circ - \theta \Rightarrow -\psi = \theta - 90^\circ$$

$$-\psi = \tan^{-1} \frac{-\zeta}{\sqrt{1-\zeta^2}} = \theta - 90^\circ = \cos^{-1} \zeta - 90^\circ$$

$$y(t) = \left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \left(\omega_n \sqrt{1-\zeta^2} t + \cos^{-1} \zeta \right) \right] u(t)$$



We could also just use Maple:

Partial fraction expansion

$$H1 := \frac{1}{s \cdot (s + a)} :$$

convert(H1, parfrac, s)

$$-\frac{1}{a(s+a)} + \frac{1}{as}$$

$$H2 := \frac{A}{s \cdot (s + a)^2} :$$

convert(H2, parfrac, s)

$$-\frac{A}{a(s+a)^2} - \frac{A}{a^2(s+a)} + \frac{A}{a^2s}$$

$$H3 := \frac{\omega_n^2}{s \cdot (s^2 + 2 \cdot \zeta \cdot \omega_n \cdot s + \omega_n^2)} :$$

convert(H3, parfrac, s)

$$\frac{-2\zeta\omega_n - s}{2\zeta\omega_n s + s^2 + \omega_n^2} + \frac{1}{s}$$

Using Maple can save you a lot of time.
Which you then can use to figure out if
the result is identical to alternative forms
😊

$$y(t) = \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin \left(\omega_n \sqrt{1 - \zeta^2} t + \cos^{-1} \zeta \right) \right] u(t)$$

restart

with(inttrans) : ←

Inverse Laplace transformation

invlaplace(H1, s, t)

$$\frac{1 - e^{-a t}}{a}$$

invlaplace(H2, s, t)

$$\frac{(-(a t + 1) e^{-a t} + 1) A}{a^2}$$

invlaplace(H3, s, t)

$$1 - \frac{1}{\omega_n (\zeta^2 - 1)} \left(e^{-t \zeta \omega_n} \left(\cosh \left(t \sqrt{\omega_n^2 (\zeta^2 - 1)} \right) \omega_n (\zeta^2 - 1) \right. \right. \\ \left. \left. + \sqrt{\omega_n^2 (\zeta^2 - 1)} \zeta \sinh \left(t \sqrt{\omega_n^2 (\zeta^2 - 1)} \right) \right) \right)$$

Lathi
6.2

Laplace theorems

Video 3

Time shift:

In the unilateral Laplace transformation, time advance is not allowed since it violates causality.

Time advance is allowed in **bilateral Laplace**.

$$\begin{aligned}
 \int_{0^-}^{\infty} x(t - t_0)u(t - t_0)e^{-st}dt &= \int_{-t_0}^{\infty} x(y)u(y)e^{-s(y+t_0)}dy & y \stackrel{\text{def}}{=} t - t_0 \\
 &= \int_0^{\infty} x(y)e^{-s(y+t_0)}dy \\
 &= e^{-st_0} \int_0^{\infty} x(y)e^{-sy}dy \\
 &= X(s)e^{-st_0}
 \end{aligned}$$

$$x(t - t_0) \leftrightarrow e^{-st_0}X(s), t_0 \geq 0$$

Frequency shift theorem:

$$\int_{0^-}^{\infty} [x(t)e^{s_0 t}]e^{-st} dt = \int_{0^-}^{\infty} x(t)e^{-(s-s_0)t} dt \\ = X(s - s_0)$$

$$x(t)e^{s_0 t} \leftrightarrow X(s - s_0)$$

Integration by parts:

$$\int f \cdot g' \, dx = f \cdot g - \int f' \cdot g \, dx$$

Differentiation theorem:

$$\begin{aligned} \int_{0-}^{\infty} \frac{dx}{dt} e^{-st} dt &= x(t) e^{-st} \Big|_{0-}^{\infty} - (-s) \int_{0-}^{\infty} x(t) e^{-st} dt \\ &= -x(0_-) + sX(s) \end{aligned}$$

$$\frac{dx}{dt} \leftrightarrow sX(s) - x(0_-)$$

$$\frac{d^2x}{dt^2} \leftrightarrow s^2X(s) - sx(0_-) - \dot{x}(0_-)$$

Examples:

$$i_C(t) = C \frac{dv_C}{dt} \qquad I_C(s) = sCV_C(s) - Cv_C(0_-)$$

$$v_L(t) = L \frac{di_L}{dt} \qquad V_L(s) = sLI_L(s) - Li_L(0_-)$$

Consider:

$$g(t) \stackrel{\text{def}}{=} \int_{0-}^t x(\tau) d\tau \quad \text{so that} \quad \frac{dg}{dt} = x(t) \quad \text{and} \quad g(0-) = 0$$

Transform pairs:

$$g(t) \leftrightarrow G(s) \quad \text{and} \quad x(t) \leftrightarrow X(s)$$

$$x(t) = \frac{dg}{dt} \leftrightarrow X(s) = sG(s) - g(0-) = sG(s)$$

$$G(s) = \frac{X(s)}{s}$$

Integration theorem 1:

$$\int_{0-}^t x(\tau) d\tau \leftrightarrow \frac{X(s)}{s}$$

Consider:

$$\mathcal{L}\{A\} = \mathcal{L}\{A u(t)\} = \frac{A}{s}$$

$$g(t) \stackrel{\text{def}}{=} \int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^{0_-} x(\tau) d\tau + \int_{0_-}^t x(\tau) d\tau = g(0_-) + \int_{0_-}^t x(\tau) d\tau$$

$$x(t) = \frac{dg}{dt} \leftrightarrow X(s) = sG(s) - g(0_-)$$

$$G(s) = \frac{X(s)}{s} + \frac{\int_{-\infty}^{0_-} x(\tau) d\tau}{s} \Leftrightarrow g(t) \leftrightarrow \frac{X(s)}{s} + \frac{\int_{-\infty}^{0_-} x(\tau) d\tau}{s}$$

Integration theorem 2:

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{X(s)}{s} + \frac{\int_{-\infty}^{0_-} x(\tau) d\tau}{s}$$

Example:

$$v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau \leftrightarrow V_C(s) = \frac{I_C(s)}{sC} + \frac{v_C(0_-)}{s}$$

Time scaling:

$$y \stackrel{\text{def}}{=} at \Rightarrow dt = dy/a$$

$$\begin{aligned} \int_{0^-}^{\infty} x(at)e^{-st}dt &= \frac{1}{a} \int_{0^-}^{\infty} x(y)e^{-sy/a}dy \\ &= \frac{1}{a} X\left(\frac{s}{a}\right) \end{aligned}$$

Time inversion is not permissible as this would turn a causal signal into an anti-causal.

$$x(at) \leftrightarrow \frac{1}{a} X\left(\frac{s}{a}\right), a > 0$$

Convolution theorem:

$$x_1(t) * x_2(t) \leftrightarrow X_1(s) \cdot X_2(s)$$

$$\begin{aligned} x_1(t) * x_2(t) &\leftrightarrow \int_{0^-}^{\infty} \int_{0^-}^{\infty} x_1(\tau) x_2(t - \tau) d\tau e^{-st} dt = \int_{0^-}^{\infty} x_1(\tau) \int_{0^-}^{\infty} x_2(t - \tau) e^{-st} dt d\tau \\ &= \int_{0^-}^{\infty} x_1(\tau) X_2(s) e^{-s\tau} d\tau = X_2(s) \cdot \int_{0^-}^{\infty} x_1(\tau) e^{-s\tau} d\tau \\ &= X_1(s) \cdot X_2(s) \end{aligned}$$

Example:

$$\begin{array}{ccccc} y_{zs}(t) & = & h(t) & * & x(t) \\ \uparrow_3 & & \downarrow_1 & & \downarrow_2 \\ Y_{zs}(s) & = & H(s) & \cdot & X(s) \end{array}$$

Let us find the Laplace transform of a product of two signals:

$$y(t) = h(t)x(t)$$

$$y(t) \leftrightarrow Y(s)$$

$$h(t) \leftrightarrow H(s)$$

$$x(t) \leftrightarrow X(s)$$

$$\mathcal{L}\{y(t)\} = \int_0^{\infty} y(t)e^{-st} dt = \int_0^{\infty} h(t)x(t)e^{-st} dt$$

$$= \int_0^{\infty} h(t) \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(p)e^{pt} dp e^{-st} dt$$

$$= \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(p) \int_0^{\infty} h(t)e^{pt}e^{-st} dt dp$$

$$= \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(p) \int_0^{\infty} h(t)e^{-(s-p)t} dt dp$$

$$= \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(p)H(s-p) dp$$

$$= \frac{1}{2\pi j} [X(s) * H(s)]$$

$$h(t)x(t) \leftrightarrow \frac{1}{2\pi j} [X(s) * H(s)]$$

Initial Value Theorem

$$x(0_+) = \lim_{s \rightarrow \infty} s X(s)$$

If the limit on the right-hand side exists.

Proof:

$$\begin{aligned} sX(s) - \cancel{x(0_-)} &= \int_{0_-}^{\infty} \frac{dx}{dt} e^{-st} dt = \int_{0_-}^{0_+} \frac{dx}{dt} e^{-st} dt + \int_{0_+}^{\infty} \frac{dx}{dt} e^{-st} dt \\ &= x(t) \Big|_{0_-}^{0_+} + \int_{0_+}^{\infty} \frac{dx}{dt} e^{-st} dt = x(0_+) - \cancel{x(0_-)} + \int_{0_+}^{\infty} \frac{dx}{dt} e^{-st} dt \\ \lim_{s \rightarrow \infty} s X(s) &= x(0_+) + \lim_{s \rightarrow \infty} \int_{0_+}^{\infty} \frac{dx}{dt} e^{-st} dt = x(0_+) + \int_{0_+}^{\infty} \frac{dx}{dt} \left(\lim_{s \rightarrow \infty} \cancel{e^{-st}} \right) dt \\ &= x(0_+) \end{aligned}$$

Final Value Theorem

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} s X(s)$$

Provided that $sX(s)$ only has poles in the left half plane.

Proof:

$$\begin{aligned} \lim_{s \rightarrow 0} [sX(s) - x(0_-)] &= \lim_{s \rightarrow 0} \int_{0_-}^{\infty} \frac{dx}{dt} e^{-st} dt = \int_{0_-}^{\infty} \frac{dx}{dt} \lim_{s \rightarrow 0} e^{-st} dt = \int_{0_-}^{\infty} \frac{dx}{dt} 1 dt \\ &= x(t) \Big|_{0_-}^{\infty} = \lim_{t \rightarrow \infty} x(t) - x(0_-) \end{aligned}$$

Example: $Y_{zs}(s) = \frac{2s}{(s+2)^2}$

Initial value: $y_{zs}(0_+) = \lim_{s \rightarrow \infty} s Y_{zs}(s) = \lim_{s \rightarrow \infty} s \frac{2s}{(s+2)^2} = 2$

Final value: $\lim_{t \rightarrow \infty} y_{zs}(t) = \lim_{s \rightarrow 0} s Y_{zs}(s) = \lim_{s \rightarrow 0} s \frac{2s}{(s+2)^2} = 0$

From a Laplace transform table: $y_{zs}(t) = (2 - 4t)e^{-2t}u(t)$

This agrees with our predictions.

So, before we do the inverse Laplace transform, we can quickly get the initial value and the final value of the response. A very valuable method to check for mistakes.

Problems

Problem 6.1 (6.1-1) Laplace transform of various functions

By direct integration [Eq. (6.8b)] find the Laplace transforms and the region of convergence of the following functions.

Use Maple/Python to help with integrations.

The regions of convergence you will have to work out yourself.

(b) $t e^{-t} u(t)$

(c) $t \cos \omega_0 t u(t)$ Hint: Substitute Euler for cos

(d) $(e^{2t} - 2e^{-t}) u(t)$

(a) $u(t) - u(t - 1)$

Problem 6.3 (6.1-4): Inverse Laplace transform and partial fraction expansion

Find the inverse Laplace transform of these functions.

$$(a) \frac{2s + 5}{s^2 + 5s + 6}$$

$$(b) \frac{3s + 5}{s^2 + 4s + 13}$$

$$(c) \frac{(s + 1)^2}{s^2 - s - 6}$$

$$(d) \frac{5}{s^2(s + 2)}$$

$$(e) \frac{2s + 1}{(s + 1)(s^2 + 2s + 2)}$$

Use Maple/Python to test if partial fraction expansion is possible. If it is, use Maple/Python to obtain the inverse Laplace transform using the results of the partial fraction expansion. Check each result by comparing with table 6.1.

Problem 6.3 a Problem 6.1.4 in newest book

$$\begin{aligned} & \text{convert}\left(\frac{2 \cdot s + 5}{s^2 + 5 \cdot s + 6}, \text{parfrac}, s\right) \\ & \qquad \qquad \qquad \frac{1}{s + 2} + \frac{1}{s + 3} \\ & \text{invlaplace}\left(\frac{1}{s + 2} + \frac{1}{s + 3}, s, t\right) \\ & \qquad \qquad \qquad e^{-2t} + e^{-3t} \end{aligned}$$

Entry 5 in table 6.1

Solutions

Problem 6.1 (6.1-1) Laplace transform of various functions

By direct integration [Eq. (6.8b)] find the Laplace transforms and the region of convergence of the following functions.

Use Maple/Python to help with integrations.

The regions of convergence you will have to work out yourself.

(b) $t e^{-t} u(t)$

(c) $t \cos \omega_0 t u(t)$ Hint: Substitute Euler for cos

(d) $(e^{2t} - 2e^{-t}) u(t)$

(a) $u(t) - u(t - 1)$

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

By direct integration [Eq. (6.8b)] find the Laplace transforms and the region of convergence of the following functions.

(b) $t e^{-t} u(t)$

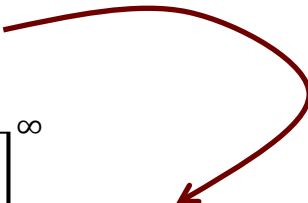
$$e^{-(s+1)t} = e^{-(\sigma+j\omega+1)t} = e^{-(\sigma+1)t} e^{-j\omega t}$$

$$|e^{-(s+1)t}| = |e^{-(\sigma+j\omega+1)t}| = |e^{-(\sigma+1)t}| \underbrace{|e^{-j\omega t}|}_{=1}$$

$$\lim_{t \rightarrow \infty} |e^{-(s+1)t}| = \lim_{t \rightarrow \infty} |e^{-(\sigma+1)t}| = \begin{cases} 0 & \text{if } \sigma + 1 > 0 \\ \infty & \text{if } \sigma + 1 < 0 \end{cases}$$

$$\begin{aligned} \int_{0-}^{\infty} t e^{-t} u(t) e^{-st} dt &= \int_{0-}^{\infty} t e^{-(s+1)t} dt \\ &= \left[-\frac{e^{-(s+1)t} (ts + t + 1)}{(s+1)^2} \right]_0^{\infty} \\ &= \frac{1}{(s+1)^2} - \lim_{t \rightarrow \infty} \frac{e^{-(s+1)t} (ts + t + 1)}{(s+1)^2} \\ &= \frac{1}{(s+1)^2}, \quad \text{Re}\{s\} > -1 \end{aligned}$$

Use Maple/Python to find the integrand



Region of convergence

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

Problem 6.1 b

Find the integrand

$$(b) \ t e^{-t} u(t)$$

$$\int t \cdot e^{-(s+1) \cdot t} dt$$

$$- \frac{e^{-(s+1) t} (s t + t + 1)}{s^2 + 2 s + 1}$$

$$\int_0^{\infty} t \cdot e^{-(s+1) \cdot t} dt$$

$$\lim_{t \rightarrow \infty} \left(- \frac{e^{-(s+1) t} s t + t e^{-(s+1) t} + e^{-(s+1) t} - 1}{s^2 + 2 s + 1} \right)$$

Inverse Laplace transform

$$\text{laplace}(t \cdot e^{-t} \cdot \text{Heaviside}(t), t, s)$$

$$\frac{1}{(s+1)^2}$$

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

By direct integration [Eq. (6.8b)] find the Laplace transforms and the region of convergence of the following functions.

(c) $t \cos \omega_0 t \, u(t)$

$$(c) \, t \cos \omega_0 t \, u(t) = t \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) u(t)$$

$$\int_{0-}^{\infty} t \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) e^{-st} dt = \int_{0-}^{\infty} t \frac{1}{2} (e^{(j\omega_0 - s)t} + e^{-(j\omega_0 + s)t}) dt$$

$$\int t e^{at} dt = \frac{(at - 1)e^{at}}{a^2}$$

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

(c) continued

$$\int_{0-}^{\infty} t \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) e^{-st} dt = \left[\frac{(j\omega_0 - s)t - 1}{2(j\omega_0 - s)^2} e^{(j\omega_0 - s)t} \right]_0^{\infty} + \left[\frac{-(j\omega_0 + s)t - 1}{2(j\omega_0 + s)^2} e^{-(j\omega_0 + s)t} \right]_0^{\infty}$$

$$\left[\frac{(j\omega_0 - s)t - 1}{2(j\omega_0 - s)^2} e^{(j\omega_0 - s)t} \right]_0^{\infty} = \frac{1}{2(j\omega_0 - s)^2}, \quad \text{Re}\{j\omega_0 - s\} < 0 \Rightarrow \text{Re}\{s\} > 0$$

$$\left[\frac{-(j\omega_0 + s)t - 1}{2(j\omega_0 + s)^2} e^{-(j\omega_0 + s)t} \right]_0^{\infty} = \frac{1}{2(j\omega_0 + s)^2}, \quad \text{Re}\{j\omega_0 + s\} > 0 \Rightarrow \text{Re}\{s\} > 0$$

$$\int_{0-}^{\infty} t \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) e^{-st} dt = \frac{1}{2(j\omega_0 - s)^2} + \frac{1}{2(j\omega_0 + s)^2} = -\frac{s^2 - \omega_0^2}{(\omega_0^2 + 2j\omega_0 s - s^2)(-\omega_0^2 + 2j\omega_0 s + s^2)}$$

$$= -\frac{s^2 - \omega_0^2}{-s^4 - 2\omega_0^2 s^2 - \omega_0^4} = \frac{s^2 - \omega_0^2}{(s^2 + \omega_0^2)^2}, \quad \text{Re}\{s\} > 0$$

$$\int t e^{at} dt = \frac{(at - 1)e^{at}}{a^2}$$

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

Problem 6.1 c

Find the integrand

$$(c) t \cos \omega_0 t u(t)$$

$$\int t \cdot \frac{1}{2} \cdot e^{a \cdot t} dt$$

$$\frac{(a t - 1) e^{a t}}{2 a^2}$$

$$\text{normal}\left(\frac{1}{\text{expand}\left(2 \cdot (j \cdot \omega_0 - s)^2\right)} + \frac{1}{\text{expand}\left(2 \cdot (j \cdot \omega_0 + s)^2\right)}\right)$$

$$- \frac{s^2 - \omega_0^2}{\left(2 I \omega_0 s - s^2 + \omega_0^2\right) \left(-\omega_0^2 + 2 I \omega_0 s + s^2\right)}$$

$$- \frac{s^2 - \omega_0^2}{\text{expand}\left(\left(\omega_0^2 + 2 I s \omega_0 - s^2\right) \left(-\omega_0^2 + 2 I s \omega_0 + s^2\right)\right)}$$

$$- \frac{s^2 - \omega_0^2}{-s^4 - 2 \omega_0^2 s^2 - \omega_0^4}$$

Inverse Laplace transform

$$\text{laplace}\left(t \cos\left(\omega_0 \cdot t\right) \cdot \text{Heaviside}(t), t, s\right)$$

$$\frac{s^2 - \omega_0^2}{\left(s^2 + \omega_0^2\right)^2}$$

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

By direct integration [Eq. (6.8b)] find the Laplace transforms and the region of convergence of the following functions.

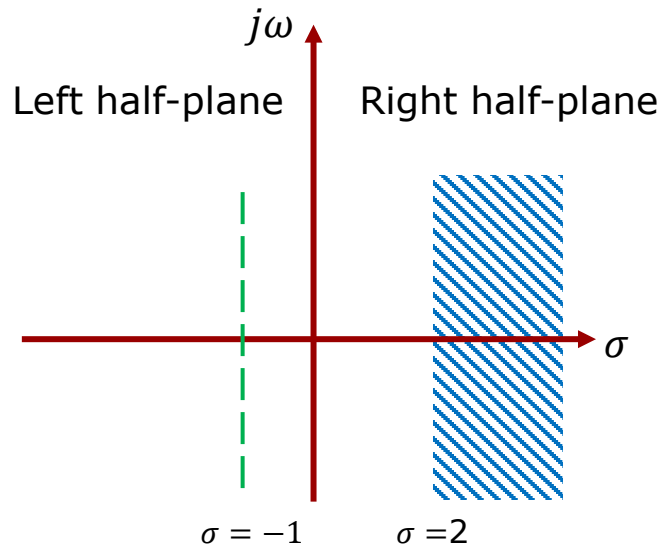
(d) $(e^{2t} - 2e^{-t}) u(t)$

$$\int_{0-}^{\infty} (e^{2t} - 2e^{-t}) e^{-st} dt = \int_{0-}^{\infty} e^{(2-s)t} dt - 2 \int_{0-}^{\infty} e^{-(s+1)t} dt$$

$$\int_{0-}^{\infty} e^{(2-s)t} dt = \left[-\frac{e^{-(s-2)t}}{s-2} \right]_0^{\infty} = \frac{1}{s-2} - \lim_{t \rightarrow \infty} \frac{e^{-(s-2)t}}{s-2} = \frac{1}{s-2}, \text{Re}\{s\} > 2$$

$$\int_{0-}^{\infty} e^{-(s+1)t} dt = \left[-\frac{e^{-(s+1)t}}{s+1} \right]_0^{\infty} = \frac{1}{s+1} - \lim_{t \rightarrow \infty} \frac{e^{-(s+1)t}}{s+1} = \frac{1}{s+1}, \text{Re}\{s\} > -1$$

$$\int_{0-}^{\infty} (e^{2t} - 2e^{-t}) e^{-st} dt = \frac{1}{s-2} - \frac{2}{s+1}, \text{Re}\{s\} > 2$$



Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

Problem 6.1 d

Find the integrand

$$(d) (e^{2t} - 2e^{-t}) u(t)$$

$$\int e^{(2-s) \cdot t} dt$$

$$= \frac{e^{-(2-s) t}}{-2 + s}$$

$$\int e^{-(s+1) \cdot t} dt$$

$$= \frac{e^{-(s+1) t}}{s + 1}$$

$$\text{laplace}((e^{2 \cdot t} - 2 \cdot e^{-t}) \cdot \text{Heaviside}(t), t, s)$$

$$= \frac{-5 + s}{(-2 + s) (s + 1)}$$

$$\text{convert}\left(-\frac{-5 + s}{(-2 + s) (s + 1)}, \text{parfrac}, s \right)$$

$$\frac{1}{-2 + s} - \frac{2}{s + 1}$$

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

By direct integration [Eq. (6.8b)] find the Laplace transforms and the region of convergence of the following functions.

(a) $u(t) - u(t - 1)$

$$\begin{aligned}\int_{0-}^{\infty} (u(t) - u(t - 1))e^{-st} dt &= \int_{0-}^1 e^{-st} dt \\ &= \left[-\frac{1}{s} e^{-st} \right]_0^1 = -\frac{1}{s} e^{-s} + \frac{1}{s} \\ &= \frac{1}{s} (1 - e^{-s})\end{aligned}$$

Time-limited signals always have an ROC equal to the entire s-plane

Problem 6.1 (6.1-1) Laplace transform of various functions (sol)

Problem 6.1 a

Find the integrand

$$(a) u(t) - u(t - 1)$$

$$\int_0^{\infty} (v(t) - v(t - 1)) \cdot e^{-s \cdot t} dt$$

$$\lim_{t \rightarrow \infty} \left(\frac{e^{-s \cdot t} \text{Heaviside}(t - 1)}{s} - \frac{\text{Heaviside}(t - 1) e^{-s}}{s} + t \text{Heaviside}(t) \right)$$

$$\text{laplace}(v(t) - v(t - 1), t, s)$$

$$\frac{1 - e^{-s}}{s}$$

Problem 6.3 (6.1-4): Inverse Laplace transform and partial fraction expansion

Find the inverse Laplace transform of these functions.

$$(a) \frac{2s + 5}{s^2 + 5s + 6}$$

Use Maple to test if partial fraction expansion is possible. If it is, use Maple to obtain the inverse Laplace transform using the results of the partial fraction expansion. Check each result by comparing with table 6.1.

$$(b) \frac{3s + 5}{s^2 + 4s + 13}$$

Problem 6.3 a Problem 6.1.4 in latest book

$$(c) \frac{(s + 1)^2}{s^2 - s - 6}$$

$$(d) \frac{5}{s^2(s + 2)}$$

$$(e) \frac{2s + 1}{(s + 1)(s^2 + 2s + 2)}$$

$$\begin{aligned} & \text{convert}\left(\frac{2 \cdot s + 5}{s^2 + 5 \cdot s + 6}, \text{parfrac}, s\right) \\ & \qquad \qquad \qquad \frac{1}{s + 2} + \frac{1}{s + 3} \\ & \text{invlaplace}\left(\frac{1}{s + 2} + \frac{1}{s + 3}, s, t\right) \\ & \qquad \qquad \qquad e^{-2t} + e^{-3t} \end{aligned}$$

Entry 5 in table 6.1

Problem 6.3 (6.1-3): Inverse Laplace transform and partial fraction expansion (sol)

Find the inverse Laplace transform of these functions.

$$(b) \frac{3s + 5}{s^2 + 4s + 13}$$

Problem 6.3 b Problem 6.1.4 in latest book

$$\begin{aligned} & \text{convert}\left(\frac{3 \cdot s + 5}{s^2 + 4 \cdot s + 13}, \text{parfrac}, s\right) \\ & \qquad \qquad \qquad \frac{3 s + 5}{s^2 + 4 s + 13} \\ & \text{invlaplace}\left(\frac{3 s + 5}{s^2 + 4 s + 13}, s, t\right) \\ & \qquad \qquad \qquad \frac{e^{-2 t} (9 \cos(3 t) - \sin(3 t))}{3} \end{aligned}$$

Entry 10d in table 6.1

Problem 6.3 (6.1-3): Inverse Laplace transform and partial fraction expansion (sol)

Find the inverse Laplace transform of these functions.

Problem 6.3 c

Problem 6.1.4 in latest book

$$(c) \frac{(s+1)^2}{s^2 - s - 6}$$

$$\begin{aligned} & \text{convert}\left(\frac{(s+1)^2}{s^2 - s - 6}, \text{parfrac}, s\right) \\ & 1 + \frac{16}{5(s-3)} - \frac{1}{5(s+2)} \\ & \text{invlaplace}\left(1 + \frac{16}{5(s-3)} - \frac{1}{5(s+2)}, s, t\right) \\ & \text{Dirac}(t) + \frac{16 e^{3t}}{5} - \frac{e^{-2t}}{5} \end{aligned}$$

Entry 1+5 in table 6.1

Problem 6.3 (6.1-3): Inverse Laplace transform and partial fraction expansion (sol)

Find the inverse Laplace transform of these functions.

$$(d) \frac{5}{s^2(s+2)}$$

Problem 6.3 d Problem 6.1.4 in latest book

$$\text{convert}\left(\frac{5}{s^2 \cdot (s+2)}, \text{parfrac}, s\right)$$

$$\frac{5}{4(s+2)} + \frac{5}{2s^2} - \frac{5}{4s}$$

$$\text{invlaplace}\left(\frac{5}{4(s+2)} + \frac{5}{2s^2} - \frac{5}{4s}, s, t\right)$$

$$\frac{5e^{-2t}}{4} + \frac{5t}{2} - \frac{5}{4} \quad \cdot u(t) \text{ implicit}$$

Entry 2+3+5 in table 6.1

Problem 6.3 (6.1-3): Inverse Laplace transform and partial fraction expansion (sol)

Find the inverse Laplace transform of these functions.

$$(e) \frac{2s + 1}{(s + 1)(s^2 + 2s + 2)}$$

Problem 6.3 e Problem 6.1.4 in latest book

$$\begin{aligned} & \text{convert} \left(\frac{2 \cdot s + 1}{(s + 1) \cdot (s^2 + 2 \cdot s + 2)}, \text{parfrac}, s \right) \\ & \quad \frac{s + 3}{s^2 + 2s + 2} - \frac{1}{s + 1} \\ & \text{invlaplace} \left(\frac{s + 3}{s^2 + 2s + 2} - \frac{1}{s + 1}, s, t \right) \\ & \quad e^{-t} (\cos(t) + 2 \sin(t) - 1) \end{aligned}$$

Entry 5 + 10d in table 6.1